

Down syndrome

Down syndrome, which affects both physical and mental development, results when an extra copy of a chromosome is made randomly due to abnormal cell division. Babies born with this disorder have identifiable facial characteristics and developmental delays evident from early infancy.

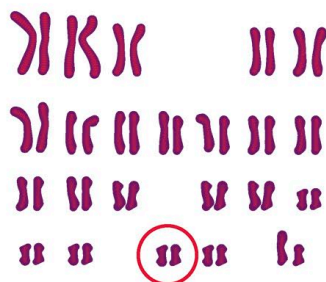
Down syndrome is also known as trisomy 21, because it creates a third copy of chromosome 21. Experiments conducted on mice have shown that the presence of this extra chromosome disrupts the function of brain circuits involved in memory and learning, mainly in the area of the hippocampus.

The chances of Down syndrome occurring in a child increase with the mother's age at pregnancy.

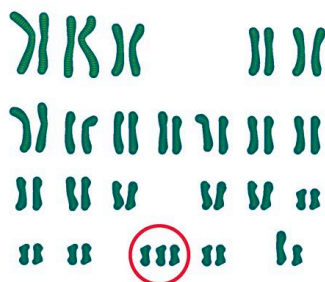
Normal and trisomy 21 chromosome sets

Two karyotypes, or photographs of a full set of chromosomes, show a normal male with two copies of chromosome 21 and a male with Down syndrome, who has three.

Everyone born with it has some level of learning disability. Certain health conditions, such as heart conditions and hearing and vision problems, are more common in people with Down syndrome.



NORMAL CHROMOSOME SET



TRISOMY 21 CHROMOSOME SET

SCREENING TESTS

Prenatal screening tests such as blood tests and ultrasounds help predict whether a child might be at risk of Down syndrome. If the risk is high, these can be followed by two diagnostic tests: chorionic villus sampling and amniocentesis, which analyze fetal cells and amniotic fluid to detect chromosome abnormalities.

Cerebral palsy

Cerebral palsy, or CP, refers to a group of disorders that impair movement, coordination, and cognition. CP is the most common childhood motor disability and is broadly defined as being either congenital or acquired.

Most children are diagnosed with congenital CP, which occurs either before or during birth as a result of brain injury, such as a difficult delivery that deprives the brain of oxygen. Brain infections or a serious head injury, however, can also cause acquired CP more than 28 days after birth.

The nature of CP symptoms depends on the location of the brain damage, but the damage is typically located in the motor cortex, which controls movement.

Symptoms and severity vary enormously and become more evident as a baby develops. Many signs of CP are often not even noticeable in newborns.

Some children with CP have impaired mobility, speech, and intellectual abilities and may require a wheelchair or need support with daily activities. Others may be floppy or rigid, have weak limbs, or have trouble walking. Depending on CP type and treatment, affected people live for 30 to 70 years.

TYPES OF CEREBRAL PALSY

CP is categorized by the movement disorder involved. A few types are listed below.

Spastic (or diplegic) CP

People with this type are very stiff and cannot relax their limbs and muscles. They may walk on their toes or with legs turned inward.

Athetoid (or dyskinetic) CP

People with this form cannot control various parts of their body and make involuntary writhing or jerking movements.

Ataxic cerebral palsy

Balance and coordination are affected, and there is often loss of voluntary muscle control when using fine motor skills such as writing.

Mixed cerebral palsy

Mixed cerebral palsy involves a combination of symptoms of CP types, due to several damaged motor-control centers in the brain.